

Multiple Choice (10 marks)

1. Which compound does not contain individual molecules?
 - (a) salt
 - (b) water
 - (c) hydrogen gas
 - (d) carbon monoxide
2. Why does a floor covered with tile feel colder than a floor covered with carpet?
 - (a) The tile floor is colder than the carpet.
 - (b) The tile floor conducts heat better than the carpet.
 - (c) The carpet conducts heat better than the tile floor.
 - (d) The tile floor is a better insulator than the carpet.
3. Genes are unable to determine a person's
 - (a) eye color.
 - (b) athletic ability.
 - (c) number of teeth.
 - (d) shape of earlobes.
4. A red giant star differs from main sequence stars because it
 - (a) burns at a cooler temperature.
 - (b) is always larger than other suns.
 - (c) has greater density and gravity.
 - (d) fuses hydrogen to make helium.
5. Which is a natural cause of rivers drying up?
 - (a) pollution
 - (b) erosion
 - (c) water tables rise
 - (d) lack of precipitation

True / False (10 marks)

Answer True or False

6. True or False: The outer layers of astronauts' space suits are reflective to protect them from intense sunlight in space.
7. True or False: Deltas forming at the mouth of rivers represent a sudden change to Earth's surface due to rapid sediment deposition.
8. True or False: Humans have the least control over the amount of water vapor present in the atmosphere.
9. True or False: Placing the coin in a freezer will lower its temperature more than any other method.
10. True or False: Complete metamorphosis describes insects that develop directly into adults without any larval stages.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the characteristics and advantages of an alternative energy resource that can be utilized similarly to coal. How does this resource impact energy production and the environment?
12. Discuss the key differences between organisms classified as Eukarya and those classified as Archaea and Bacteria, focusing on their cellular structures and functions.

End of Paper